# BEYOND EVALUATION USE

## An Empirical Study of Involvement and Use/Influence in Multi-Site Evaluations

**Report 8:**
**Cross Case Report**

University of Minnesota
College of Education and Human Development
Minneapolis, MN 55455

Last Updated March 2009

*Note:* This material is based upon work supported by the National Science Foundation under Grant No. REC 0438545. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the authors and do not necessarily reflect the views of the National Science Foundation.

# Contributors

Principal Authors.....Frances Lawrenz
.....Jean A. King
Principal Investigators.....Frances Lawrenz
.....Jean A. King
Research Assistants (alphabetical) .....Lija Greenseid
.....Gina Johnson
..... Kelli Johnson
.....Ann Ooms
.....Denise Roseland
..... Stacie Toal
..... Boris Volkov

## Table of Contents

**Executive Summary** ..... **1**

**Introduction** ..... **3**

**Methods** ..... **4**

**Overview of the Four Program Evaluation Projects** ..... **6**

ATE Program Overview ..... 6

ATE Program Evaluation Project ..... 7

CETP Program Overview ..... 9

CETP Program Evaluation Project ..... 9

LSC Program Overview ..... 12

LSC Program Evaluation Project Overview ..... 12

MSP-RETA Program Overview ..... 14

MSP-RETA Program Evaluation Project Overview ..... 15

**Cross-Case Analyses** ..... **17**

Perceptions of Evaluation Quality ..... 17

Interface with NSF ..... 18

Life Cycles ..... 19

Project Control ..... 21

Community and Networking ..... 22

Tensions ..... 24

Unique Aspects ..... 25

**Conclusions** ..... **25**

# Executive Summary

This cross-case analysis is part of a series of reports examining involvement and use of evaluation processes and outcomes in multi-site settings. The reports are part of the Beyond Evaluation Use research project funded by the National Science Foundation (NSF) from 2004-2009 to examine involvement in and use of evaluation processes and outcomes in four different, large, multi-site NSF programs. The research team also investigated the influence of these evaluations on the fields of science, technology, engineering, and mathematics (STEM) education and evaluation.

This study focused on the evaluations of four NSF programs: the Advanced Technology Education Program (ATE evaluation project), the Collaboratives for Excellence in Teacher Preparation Program (CETP evaluation project), the Local Systemic Change Program (LSC evaluation project), and the evaluation capacity building for the Mathematics and Science Partnerships Program (MSP-RETA evaluation project). The research team prepared detailed case study reports examining involvement and use in each of the four evaluation projects. This cross-case analysis highlights similarities and uniqueness across the cases about the relationship between involvement in evaluation and the use of the evaluation by people other than the primary intended users (NSF). The cases were selected specifically to represent a continuum of project control over the evaluation efforts.

To best understand the results of our research, it is helpful to understand that this research is different than other research about evaluation use in three important ways. The first major difference is that this research was not about an evaluation in a single place or with a finite group of people. Instead, this research addressed large multi-site evaluations where the "participants" were defined not as individual people but rather as individual projects, each implementing a different approach to solving a national problem at different sites across the country. To highlight this distinction, we used the term "involvement" rather than participation (Lawrenz & Huffman, 2003; Toal, King, Lawrenz, & Johnson, 2009; Toal, in press).

Second, this research focused on the use of evaluation by secondary, somewhat unintended, users. Although NSF was the primary user and client in each of the four program evaluations, the focus was not on NSF, but rather on the local projects and on the fields of STEM education and evaluation, regardless of whether the national program evaluators were instructed to consider project and field use in the design of the program evaluation.

The third and final distinction is that in addition to providing NSF with information for decision-making, these four large program evaluations also carried the traditional NSF scientific knowledge-generating purpose that moved them more into the realm of research.

The results of this cross case study suggest several ideas for affecting involvement and subsequent use of evaluation processes and outcomes by unintended audiences. The most

important aspect of facilitating involvement and use appears to be careful and continuous communication by the evaluation team to the participating projects as well as the field through a variety of mediums. It is important to have a clear interface with the direct user of the evaluations because this relationship affects how the individual projects and the field interact with the evaluation. Evaluators need to be cognizant of the life cycles of the individuals and the projects involved in the evaluation and plan for their different types of involvement and use. Tensions between the overall evaluation and the local projects and their evaluation efforts should be expected. A seemingly fruitful way to use evaluation funds is to develop a sense of community among the local projects through face-to-face meetings as well as other types of contact. Developing a community of practice with mutual respect and true interest in each other's ideas as well as reasonable expectations for participation are also important.

# Introduction

This cross-case analysis is part of a series of reports examining involvement and use of evaluation processes and outcomes in multi-site settings. The reports are part of the Beyond Evaluation Use research project funded by the National Science Foundation (NSF) from 2004-2009 to examine involvement in and use of evaluation processes and outcomes in four different, large, multi-site NSF programs. The research team also investigated the influence of these evaluations on the fields of science, technology, engineering, and mathematics (STEM) education and evaluation.

This study focused on the evaluations of four NSF programs: the Advanced Technology Education Program (ATE evaluation project), the Collaboratives for Excellence in Teacher Preparation Program (CETP evaluation project), the Local Systemic Change Program (LSC evaluation project), and the evaluation capacity building for the Mathematics and Science Partnerships Program (MSP-RETA evaluation project). The research team prepared detailed case study reports examining involvement and use in each of the four evaluation projects. The purpose of any cross-case analysis is to highlight similarities and uniqueness across the cases in order to shed light on the questions or descriptions of interest. The question being examined here was the relationship between involvement in evaluation and the use of the evaluation by people other than the primary intended users (NSF). The cases were selected specifically to represent a continuum of project control over the evaluation efforts. Therefore, one of the projects, the one at the higher project control end of the continuum, was quite different from the other three. The outlier, MSP-RETA, provided evaluation technical assistance at the requests of the projects in the MSP program while the other three cases were program evaluations with varying degrees of project control over the program evaluation. Each evaluation project operated on a unique timeline for differing durations, with the ATE evaluation project still ongoing in 2009 and the CETP program evaluation running for only 4 years. Additionally, although all projects were about STEM education, each evaluation project examined different aspects of it, ranging from mostly community college-based STEM workforce development, to district based in-service teacher professional development, collaboratives for pre-service teacher education or partnerships encompassing multiple aspects of STEM education.

To best understand the results of our research, it is first helpful to understand that our project operated in a unique space, crossing boundaries that have constrained other evaluation research. Consequently, this research is different than other research about evaluation use in three important ways. The first major difference is that this research was not about participatory evaluation as commonly understood in the evaluation field, i.e., an evaluation in a single place or with a finite group of people. Instead, this research addressed large multi-site evaluations where the “participants” were defined not as individual people but rather as individual projects, each implementing a different approach to solving a national problem at different sites across the country. Therefore, the study collected data from individual people

serving as representatives of their projects. To highlight this distinction, we used the term "involvement" rather than participation (Lawrenz & Huffman, 2003; Toal, King, Lawrenz, & Johnson, 2009; Toal, in press).

Second, this research focused on the use of evaluation by secondary, somewhat unintended, users. Although NSF was the primary user and client in each of the four program evaluations, the focus was not on NSF, but rather on the local projects and on the fields of STEM education and evaluation, regardless of whether the national program evaluators were instructed to consider project and field use in the design of the program evaluation.

The third and final distinction is that in addition to providing NSF with information for decision-making, these four large program evaluations also carried the traditional scientific knowledge-generating purpose of evaluation that moved them more into the realm of research. Furthermore, NSF expects that the knowledge generated by all of its projects will be disseminated to the public. Although it is not exactly evaluation use, the dissemination of the knowledge generated by NSF projects is targeted at other projects and the STEM field in the hope that this knowledge will be useful. Thus, the four evaluation projects being researched are playing somewhat dual roles: first, as research projects that disseminate knowledge to other projects and the STEM field and second, as evaluation projects that serve the needs of their client, NSF. This makes the notion of "evaluation use" much more complex. The Beyond Evaluation Use research project investigated the effect of involvement in the evaluations on the use of the evaluations' processes and outcomes on secondary users—the projects participating in the evaluations and the STEM field.

The next section in this report describes the methods used in the Beyond Evaluation Use Research Project. Following the methods is a summary of each of the multi-site NSF programs and their evaluations. The final two sections of this report discuss the themes related to involvement and use that were found across the cases and conclusions that may be drawn from this cross-case analysis.

## Methods

A variety of data collection methods were used to construct the detailed case studies of the four evaluation projects and to provide the basis for the other reports that are part of this research project. Data collection methods included: surveys, interviews, evaluator reflection, archival review and citation analysis.

An online survey about involvement in and use of the processes and outcomes of the four evaluation projects was distributed to 935 individual project PIs, co-PIs, coordinators, directors, project leaders and evaluators connected to the four national evaluation projects. The overall response rate was 46%. A non-respondent follow-up survey was administered to a random selection of non-responding project leaders and evaluators. The results of the non-respondent

study showed no significant difference in the reported overall level of involvement for each of the four projects. However, there was a significant difference between respondents and non-respondents in the ATE and CETP evaluation projects regarding the impact of the evaluations. In both cases the respondents reported higher levels of impact than non-respondents.

In-depth interviews were conducted with two purposefully selected samples of project leaders and evaluators who had responded to the survey. One sample contained respondents with varying reported levels of involvement and use, e.g., low use—high involvement, while the other sample was entirely high users. Twenty-nine interviews were conducted with about eight interviews per evaluation project. Interviews lasted between 30-50 minutes. All were taped and transcribed. Interviews were also conducted with one of the NSF program officers involved with each of the four projects.

The four evaluators who directed the evaluation projects provided in-depth reflections on their experiences through written exercises, two advisory committee meetings, and a day-long retreat. The evaluators were asked to reflect on the intended and actual levels of individual project involvement in their evaluation projects and how the individual projects and others used the program evaluation findings and processes.

An online survey of the field was administered to a sample of 350 individuals randomly selected from the database of over 2000 currently-funded NSF Education and Human Development Directorate PIs. The survey elicited responses about PIs' perceptions of which evaluations might have been influential on the fields of STEM evaluation and education. The overall response rate was 54.7%.

A very brief email/telephone survey of editors and coeditors of six journals in the evaluation and STEM education fields was also conducted to determine if the editors were aware of any articles that had been published about influential STEM evaluations. Responses were obtained from three of six editors or coeditors.

Data were also obtained through a review of archival documents, publications, and reports produced by the four evaluation projects. Archival materials were obtained through Web searches and through contact with the evaluation project PIs.

A citation analysis was conducted using a complete list of all publications, presentations, and instruments developed by the four evaluation projects to analyze patterns of influence the evaluations had on the fields of STEM education and evaluation. There were 246 project artifacts identified and 376 citations.

# Overview of the Four Program Evaluation Projects

Although detailed information on each of the evaluation projects is provided in the individual case studies, overview information is provided here to provide a context for the cross-case analysis. The time line below outlines how the four evaluation projects fit into history.

[INSERT TIMELINE HERE]

**ATE Program Overview**

Working under a Congressional mandate, NSF created the Advanced Technological Education (ATE) program in 1992 to improve advanced technological education. NSF aimed to increase the number and quality of skilled technicians in the workforce thus improving U.S. competitiveness amidst growing global competition. Viewed as a conduit for preparing technicians, community colleges received most of the ATE funds to strengthen scientific/technical education and training capabilities in biotechnology, environmental technology, GIS, manufacturing, and telecommunications.

**ATE Program Facts**

**Period**: 1993-present

**Purpose**: (1) Build capacity to provide advance tech education; (2) Increase number of skilled technicians.

**Objectives:** Projects focused on improving technical education materials, enhancing technical instruction, and providing professional development to faculty and teachers. Grantees established partnerships with high schools, two- and four-year colleges, businesses, government agencies, and professional societies.

**Primary Target:** Students & teachers in 2-year colleges and collaborating institutions.

**Sites Funded:** Between 1994 and 2005, ATE funded 674 projects and centers, making grants to 345 unique institutions of which more than 200 were two-year colleges.

This congressional charge resulted in a large diversity of high-budget projects and centers. For example, between 1994 and 2005, ATE funded 674 projects and centers, making grants to 345 unique institutions of which more than 200 were two-year colleges. The cost of the program totaled approximately $350 million. ATE funded large Centers of Excellence (up to $5 million for 4 years) and smaller-scale projects (up to $300,000 for 3 years). ATE projects focused on improving technical education materials, enhancing technical instruction, and providing professional development to faculty and teachers. Grantees established partnerships with high schools, two- and four-year colleges, businesses, government agencies, and professional societies to improve the nation's advanced technology workforce.

**ATE Program Evaluation Project**

The diversity of ATE projects, in terms of scope, stakeholders, and organizations, and project or center practices created unique challenges to understanding program-level effectiveness. Nevertheless, NSF remained committed to both project-level and program-level evaluation. Therefore, in 1999, NSF funded the Evaluation Center at Western Michigan University (WMU) to conduct an external evaluation of the entire ATE program. The evaluation budget was appropriately large. Arlen Gullickson, PI, received two consecutive NSF grants: the initial grant of $1.3 million covered 1999-2002, and a subsequent $1.8 million grant covered 2002-2005. Although some components of the ATE program evaluation (primarily the annual web-based survey) are still on-going, the scope of this case study is limited to the years 1999 – 2005. Consequently, activities that have occurred since 2005 are neither considered nor commented upon, despite their potential impact on involvement and use or influence.

**ATE Evaluation Facts**

**NSF Grant Funding**

•$1.3 million (1999-2002)
•$1.8 million (2003-2005)

**NSF Goals for ATE Program Evaluation**

•Provide evidence of quality of ATE program
•Inform program improvement

**Evaluation Questions**

•To what degree is the program achieving its goals?
•Is it making an impact and reaching the individuals and groups intended?
•How effective is it when it reaches its constituents?
•Are there ways to significantly improve the program?

**Personnel/Expertise**

•NSF 3-person mgmt team
•Experienced evaluation team
•Evaluation Advisory Committee

The evaluation focused on program-wide productivity and accountability with the primary audience being NSF and then Congress. It is important to understand that the evaluators structured the evaluation in a manner that would allow them to provide feedback to NSF. Consequently, the evaluation was not designed to necessarily influence the projects and centers. With that said, program-level feedback was disseminated to projects in hopes of improving project-level processes and outcomes.

Principal evaluation activities included: collecting descriptive data, developing an annual web-based evaluation survey to collect data over time, conducting site visits, drafting targeted reports on various aspects of the program, and providing formative evaluation information to NSF and to the projects and centers. The annual web-based surveys collected quantitative data on the activities, accomplishments, and effectiveness of ATE projects and centers for general accountability purposes. The series of 13 extensive site visits to centers and projects validated and illuminated survey findings and allowed evaluators to gather detailed information about project operations and outcomes. The sites were selected using a purposive sampling technique based on survey data to be representative of the diversity of the ATE projects. Teams of trained

site visitors included evaluators, industry representatives, and education experts. Finally, the evaluators conducted four targeted studies on different program components: 1) the value added by ATE projects and centers to business and industry; 2) the development of materials for use by ATE projects; 3) the professional development provided by the ATE projects, and: 4) the sustainability of the impacts of ATE projects.

The projects involved in the ATE evaluation project felt that completing the survey was useful and helped them to better conduct their projects. They commented that managing the survey process helped them to gain evaluation skills and to better understand their own projects. The projects reported that they applied the evaluation project results to their own projects and that the results helped with decision plans. Additionally they reported that having the survey results helped them to meet the governmental requirements of compliance. Despite this, they reported that participating in the ATE evaluation project took a great deal of time.

The citation patterns identified in the citation analysis suggest that the products produced by the ATE program evaluation were primarily influential on other ATE sources. This does not mean, however, that the knowledge produced by the ATE evaluation was limited in its influence to a small group of evaluators engaged in its day to day work. In fact, there were 24 separate individuals who were listed as authors on one or more of the ATE evaluation products, as the ATE evaluation engaged a variety of internal staff and external consultants in its work. That said, these citation levels and patterns do suggest that the greatest influence of the ATE evaluation products was on those affiliated with the ATE national evaluation in some way, as opposed to the field more broadly.

As far as influence on the fields of STEM education and evaluation, it appears that there were a few key evaluation-related publications and workforce development publications that had the greatest influence on individuals who were not affiliated with the ATE evaluation in any direct way. The ATE evaluations' published articles and monograph were most often used externally, whereas reports were primarily cited by internal sources, and presentations were not cited in any other works.

Finally, results for the field influence survey provided a small amount of evidence suggestive of influence of the core ATE evaluation, the PIs,
and related publications on the broader fields of STEM education and evaluation. Sixty-four of 191 respondents indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. Within these 51 named evaluations, the ATE evaluation was named four times. In addition, researchers asked survey respondents to note STEM evaluation publications they viewed as influential. Only four of 44 respondents actually named a publication associated with ATE. Finally, the team asked respondents to identify influential STEM evaluators and the evaluator associated with ATE was not present in the responses.

## CETP Program Overview

In the early 1990s, NSF responded to the national need for more well-qualified math and science teachers by creating the Collaboratives for Excellence in Teacher Preparation (CETP) program. CETP sought to improve teacher preparation by enhancing science, mathematics, and educational methods courses at the college level. The program also sought to increase collaboration within and between K-12 and higher education institutions to foster the development of well-trained teachers competent in science and math.[1] As the name indicates, collaboration was the hallmark of the CETP program. To be eligible, projects had to demonstrate institutional support and collaboration among two-year, four-year, and research institutions, school districts, the business community, and state departments of education.

**CETP Program Facts**

**Period**: 1993-2004

**Purpose**: To improve teacher preparation by enhancing science, mathematics, and educational methods courses at the college level

**Objectives**: Increase collaboration within and between K-12 and higher education institutions to foster the development of well-trained teachers competent in science and math.

**Primary Target:**

**Sites Funded:** From 1993 – 2000, NSF funded 19 CETPs, each involving 3-15 higher education institutions (including colleges, universities, and community colleges) and several school districts.

From 1993 –2000, NSF funded 19 CETPs, each involving 3-15 higher education institutions (including colleges, universities, and community colleges) and several school districts. Individual projects could apply for up to $1 million per year for three to five years (system-wide-focus awards.)[2] After the initial awards, CETPs could apply for three additional years of funding, in part, to cover the costs of evaluation activities (follow-up awards.)

Initially, NSF required each CETP project to conduct its own project-level evaluation. Several years later, building upon the recommendations from two meetings of the CETP project PIs, NSF funded a national, overarching program-level evaluation effort through a $990,000 three-year grant to PI Frances Lawrenz at the University of Minnesota.

## CETP Program Evaluation Project

The CETP core evaluation goals were: (1) to achieve consensus on evaluation questions, (2) to develop, pilot, and field test evaluation instruments tied to the questions, and (3) to collect data to document CETP accomplishments nationwide.[3] NSF required the core evaluation

---

[1] Lawrenz, F., Huffman, D. & Graves, A. Impact of the Collaboratives for Excellence in Teacher Preparation Program. *Journal of Research in Science Teaching.* 2007, 44(9):1348-1369

[2] National Science Foundation. (1999). Teacher Preparation Awards: NSF Collaboratives for Excellence in Teacher Preparation Awards. Retrieved from http://www.nsf.gov/pubs/1999/nsf9996/start.htm#_Toc442862272

[3] Lawrenz, F., Appeldoorn, K., Michlin, M. (2003, July). Progress Report: CETP Core Evaluation Project. Report to NSF. Made available by CETP Principal Investigator.

services to be available to all CETP projects, but it did not require the CETP projects to participate or to share data with the Core. Because involvement was optional, the Core relied on incentives to encourage individual CETPs to participate in the core evaluation process.

The results of the survey and interviews provided support to the assertion that involvement in the planning stages of a multi-site evaluation is related in important ways to whether the evaluation process influences individual project leaders and evaluators in meaningful and lasting ways. Leaders and evaluators who were involved in the evaluation planning process reported gaining knowledge and understanding of evaluation processes as well as gaining valuable evaluation skills. Additionally, the commitment to and belief in the value of the overall program evaluation appeared to be enhanced by participation in the development of data collection instruments, while exclusion from this process may have led to some individuals questioning the quality and value of common data collected from multiple sites. The individual projects reported that the evaluation project had developed a sense of community which both supported their involvement in the evaluation project and built their evaluation capacity through use of evaluation processes and outcomes. They felt their knowledge of evaluation was increased and that they now knew to plan evaluations early. The projects felt they learned about broader issues in STEM evaluation and the problems that were facing their colleagues. Over 80% of those who did another evaluation indicated that they had used what they learned in the CETP evaluation project in another evaluation. On the other hand, some projects felt that participating in the evaluation took a great deal of time. Some felt that the instruments developed through the evaluation project didn't align with their individual project activities. Although there were some changes in beliefs about one third of the survey respondents reported no influence.

**CETP Evaluation Facts**

**NSF Grant Funding**

- $999,000 (1999-2004)

**NSF Goals for CETP Program Evaluation**

To learn to what extent the CETPs succeeded in achieving significant and systemic improvement in the science, technology, engineering, and mathematics (STEM) preparation of prospective pre-Kindergarten through grade 12 (preK-12) teachers

**Evaluation Questions**

- To what extent did the CETP program impact the collaboration and focus of university faculty on instructional issues?
- To what extent did the CETP program impact the instructional techniques used by university faculty?
- Did K-12 teachers who participated in CETP projects view their preparation programs differently from teachers who participated in other preparation programs?
- Were the instructional practices exhibited by K-12 teachers who participated in CETP projects different from the instructional practices exhibited by teachers who participated in other preparation programs?

**Evaluation Activities**

Convened meetings with CETP project personnel

- Developed data collection instruments (surveys, classroom observation protocols.)
- Provided technical assistance to local CETPs for data collection and analysis
- Standardized instruments were developed, but sites were free to use their own evaluation instruments or could add items to the standard instrument

According to some individuals interviewed and surveyed, the core evaluation did not appear to have a very large impact on the individual CETP projects or on the STEM education and evaluation fields, while for others the evaluation's influence was extensive. This variation can be explained to some extent by whether individual projects were at a point in their lifecycle to benefit from being part in the collaborative process of developing the core evaluation instruments and protocols.

The results of a citation analysis examining citations to the 24 evaluation reports, instruments, journal articles, and conference presentations produced in conjunction with the CETP core evaluation provide further examples of the influence of the core on the projects. A total of 42 citations contained within 38 unique publications were found to the 24 evaluation products. Of these, 16 of the 38 citing publications (42.1%) were authored by individuals related to one of the CETP projects. The majority of the citations (27 of 42) were to the CETP Classroom Observation Protocol - one of the instruments developed collaboratively by the core and project PIs and evaluators.

Dissemination was not a key priority for the CETP core evaluation, which was instead primarily concerned with creating an evaluation process and instruments collaboratively with the input of CETP project PIs and evaluators. That said, the evaluation instruments appear to have had some impact on the field of STEM education, at least within the area of pre-service teacher training. Among the CETP core evaluation products, the CETP Classroom Observation protocol was clearly most highly influential in terms of being cited within CETP projects as well as by unaffiliated sources. The remaining CETP products were picked up in one or two places but did not demonstrate such a high level of influence. In general it appears that the evaluation instruments seemed to have the greatest influence on the STEM fields, particularly on individuals connected to the evaluation in some way.

The research team examined the field survey results for evidence of influence of the CETP evaluation, its PIs, and related publications on the broader fields of STEM education and evaluation. The field survey results show 34% (n=64) of all respondents (n=191) indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. Within these 51 named evaluations, the CETP evaluation was only named once. In addition, researchers asked survey respondents to note STEM evaluation publications they viewed as influential. Only one of 44 respondents actually named a publication associated with CETP. Finally, the team asked respondents to identify influential STEM evaluators and three of 59 respondents named an evaluator associated with CETP.

## LSC Program Overview

NSF launched the Local Systemic Change through Teacher Enhancement initiative (LSC) in 1995 to improve K-12 science, mathematics, and technology instruction through teacher professional development. LSC intended to shift the professional development focus from individual teachers to district- or system-wide populations of teachers, based on the premise that system-wide initiatives yielded better results.[4]

**LSC Program Facts**

**Period**: 1995 – 2005 (Final year of new projects were funded was 2002.)

**Purpose**: Improve instruction in science, mathematics and technology through teacher professional development within entire school districts.

**Objectives**: Target all teachers in a jurisdiction for professional development, requiring a minimum of 130 hours of professional development; and prepare teachers to implement district-designated instructional materials.

**Primary Target:** K-12 teachers of science and mathematics; focus on entire school systems or districts, not on individual teachers.

**Sites Funded**: 88 projects current and completed projects across 31 states, and involving 70,000 teachers, 4,000 schools, 467 school districts.

The LSC was a $250 million program that funded 88 projects, reaching 4,000 schools across 31 states and 476 districts in the period 1995-2005. Some LSC projects took place in single school district and others covered multiple districts (up to 20 in some cases). Half of the schools were urban, a quarter suburban, and the remaining quarter split between small cities and rural communities. LSC-treated teachers taught approximately 70,000 students, approximately half of whom were from minority groups. The professional development sessions took place both in the summer as well as throughout the academic year, providing teachers with opportunities to develop and apply new knowledge and skills within a supportive school/district culture.[5]

## LSC Program Evaluation Project Overview

In 1995, NSF contracted with Horizon Research, Inc. (HRI) and PI Iris Weiss, Ph.D., to conduct the nationwide LSC core evaluation. NSF required LSC projects both to conduct individual project-level evaluations and to participate in the national cross-project core evaluation. The core evaluation was intended to assess the status of individual LSC projects, as well as to collect information from as many diverse projects that could be aggregated to inform broader messages to NSF, Congress and the public. NSF's theory of action was that high quality professional development focused on standards-based instructional materials would improve teachers' knowledge and skills and would increase their disposition to use those materials in their classrooms. Although somewhat modified over time as NSF's information needs changed

[4] Banilower, ER, Boyd SE, Pasley JD, and Weiss IR. Lessons from a Decade of Mathematics and Science Reform: A Capstone Report from the Local Systemic Change through Teacher Enhancement Project. Accessed at http://www.horizon-research.com/reports/2006/capstone.php

[5] Horizon Research Inc. Findings from a Decade of Local Systemic Change (LSC) Projects Web site. Retrieved September 28, 2007 from http://www.pdmathsci.net/about/on 11-29-06.

and the need for longitudinal data became apparent, the core evaluation required each project to:

- Observe 5-8 professional development sessions per year and record observations on standardized protocols;
- Administer 300 teacher questionnaires to a random sample of teachers in the participating school districts;
- Administer questionnaires to principals in all participating schools;
- Conduct classroom observations of a minimum of 16 randomly selected teachers;
- Conduct interviews with 10 randomly selected teachers;
- Interview the project administrative team; and
- Collect information on district policies and practices.

The individual LSC projects were required to participate in the extensive LSC evaluation project after the first few years. The projects reported that a sense of community was developed along with a paradigm shifting-consensus within the community about STEM teaching. Eighty-five percent reported that they used evaluation project findings to support other activities. Over 88% said they increased their evaluation knowledge, built skills and learned more about their project and about how to plan future evaluations. In contrast, the evaluation reports were viewed as extensive and difficult.

**LCS Core Evaluation Facts**

**NSF Contract Amount**

- $6.25 million

**NSF Goals for LSC Core Evaluation**

- Provide aggregated information across diverse projects to inform broader conclusions about design, quality, and impact of LSC projects
- Assess individual projects to provide for mid-course adjustments

**NSF Contracted with HRI to:**

- Develop a data collection framework
- Provide technical assistance in implementing evaluation activities
- Prepare cross-site analyses of evaluation results

**Local Project Evaluator Activities:**

- Professional Development Session Observations (2,40 completed)
- Classroom Observations of mathematics and science lessons (1,620)
- Teacher Questionnaires (75,000)
- Principal Questionnaires (17,380)
- Teacher Interviews (1,782).

The LSC citation analysis suggested that two types of products were most highly influential on the STEM fields. The first type of highly cited product is evaluation instruments such as the Classroom Observation Protocol, Core Data Collection Manual, and the surveys of teachers and administrators. The interest in these instruments can likely be attributed to two factors. First, the instruments filled a need within the field; there was a lack of high-quality instruments for measuring outcomes of interest in mathematics and science teacher professional development programs. Second, the LSC program evaluation involved data collection from 88 LSC projects in 467 districts across the United States. Consequently a very

high number of teachers, administrators, and evaluators were involved with assessing or being assessed by these instruments.

The second type of product that was highly cited is summary oriented reports that present the most important findings and lessons learned from the evaluation. These included such publications as the projects recently published capstone report "Lessons from a decade of mathematics and science reform" and other key evaluation reports that presented summative findings.

Although a number of evaluation instruments, reports, and publications were found to have an impact on the STEM fields, it is important to note that none of the numerous LSC Core Evaluation newsletters or presentations made by HRI staff was found to be cited in this analysis. While these dissemination efforts may have contributed indirectly to a greater knowledge of the core evaluation and its findings, these types of products appear to be relatively ineffective in contributing to a more formalized spread of knowledge as captured by citations.

The research team examined the field survey results for evidence of influence of the LSC evaluation, its PIs, and related publications on the broader fields of STEM education and evaluation. The field survey results show 34% (n=64) of all respondents (n=191) indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. Of these, the LSC evaluation was named by four respondents. Similarly, of the 44 respondents naming a publication four named a publication associated with LSC. Finally, in response to the question regarding influential STEM evaluators, seven of 59 respondents named an evaluator associated LSC – the greatest single response to this question.

**MSP-RETA Program Overview**

The National Science Foundation Math Science Partnership Program (MSP) began as a presidential initiative in 2001 and was authorized by Congress in 2002. MSP is a major research and development effort that supports innovative partnerships to improve K-12 math and science instruction, enhance student achievement, and reduce the achievement gap in math and science performance among diverse student populations. In that effort, MSP promotes efforts to develop partnerships between higher education and K-12 STEM teachers. The partnerships must support the program's goals to provide challenging coursework; increase the quantity, quality, and diversity of math and science teachers; and establish results-oriented projects that implement evidence-based educational practices. Typical project activities include new curriculum development and skill-building for teachers. Successful projects serve as models that can be widely replicated.

In 2006, MSP was bringing together 150 institutions of higher education, 550 school districts, and 3,300 schools, along with corporate partners including Pfizer, Ford, Texas Instruments, Inc., Xerox, IBM, Merck, and Intel. The MSP also created a subset of projects dedicated to research, evaluation and technical assistance (aptly titled Research, Evaluation and Technical Assistance projects). The MSP-RETAs conduct and support evaluation and research to improve K-12 STEM education.

**MSP-RETA Program Facts**

**Period**: 2002-present

**Purpose**: Research and development effort that supported innovative partnerships to improve K-12 math and science instruction, enhance student achievement, and reduce the achievement gap in math and science performance among diverse student populations.

**Objectives**: Supported the development of partnerships to integrate the work of higher education and K-12 STEM faculty through challenging coursework; Sought to increase the quantity, quality, and diversity of math and science teachers; and established results-oriented projects that implement evidence-based educational practices.

**Primary Target:** Teams of institutions: higher education, K-12, and other partners.

**Sites Funded**: In 2006, MSP was bringing together 150 institutions of higher education, 550 school districts, and 3,300 schools, along with corporate partners

The RETA projects directly support the work of the MSPs, especially by developing tools to assess teachers' growth in the knowledge of mathematics or the sciences needed for teaching, conducting longitudinal studies of teachers and their students who participate in the MSP projects, or engaging the national disciplinary and professional societies in MSP work. Of the more than 30 MSP-RETA projects, this case study considers just one: the Utah State University MSP-RETA project, which was titled "Building Evaluation Capacity of STEM Projects-MSP," established in October 2002. Unlike almost all the other MSP-RETA projects, the Utah State MSP-RETA received a grant of $1.5 million over three years, while most received only one year of funding. A no-cost NSF grant extension and an additional supplemental grant of approximately $300,000 helped continue work of the Utah State MSP-RETA for a fourth year.

**MSP-RETA Program Evaluation Project Overview**

The purpose of the Utah State MSP-RETA was not to evaluate MSP projects, per se. Rather, this RETA was funded to provide direct technical assistance to 10 MSP projects per year related to identifying evaluation needs, developing program evaluation models based on those needs, and building evaluation capacity within those projects. Further, this RETA offered technical assistance and expert consultation via a network of evaluation consultants and supported the development of professional communities of teachers, researchers, evaluators, cognitive scientists, and social scientists and others who depended on evaluation for continued improvement and for generating new knowledge. To that end, the Utah State MSP-RETA created program-wide evaluation assistance in the form of educational seminars, conferences, materials, and individualized technical assistance. Through these activities with local MSP

projects, the Utah State MSP-RETA informed and promoted evaluation at the project level but did not engage in core MSP evaluation activities or conduct evaluations of local projects.

Most of the survey data were not applicable to this evaluation project because the wording of the survey items was misleading. Some of the open-ended responses, however, were able to be included. The MSP-RETA was a dichotomous project offering direct evaluation consulting to individual projects as well as workshops for the projects at large. The expert consultants were somewhat underutilized. Despite this, the projects reported that the evaluation project promoted the development of a community. Many reported that they used the DIO model although some felt it was not applicable to their project. The individual projects also felt the evaluation project deepened their knowledge of evaluation and STEM.

**Utah State MSP-RETA Evaluation Facts**

**NSF Grant Funding**

- $1.5 million initially; additional funds of $300,000 received during year three of efforts

**NSF Goals for MSP-RETA Program Evaluation**

- Work with the MSP projects to identify evaluation needs and develop program evaluation models based on those needs
- Offered technical assistance and expert consultation via a network of evaluation consultants

**Project Activities**

- Evaluation Capacity-building seminars, conferences, materials
- Provided individualized technical assistance
- Developed and disseminated the Design-Implementation-Outcomes (DIO) Cycle framework

When conducting the citation analysis to examine the possibility of influence of the MSP-RETA, The Utah State MSP-RETA's products had limited influence on the STEM fields, with the exception of the D-I-O Model report, which has been picked up in several STEM and non-STEM areas. Interestingly, the citations to the D-I-O cycle have been primarily in collections of online resources as opposed to written reports or articles. As the D-I-O cycle report has only been published for a few years, however, it's likely that the influence of the publication will spread and will reach journals and reports as time passes. It is somewhat surprising that the D-I-O cycle has not been cited more widely within the MSP projects as it has been a focus at a number of MSP conferences, however it may be more a matter of it being hard to find references to its use as opposed to it not actually being used in evaluation planning by the MSP projects.

The research team examined the field survey results for evidence of influence of the MSP-RETA evaluation, its PIs, and related publications on the broader fields of STEM education and evaluation. The field survey results show 34% (n=64) of all respondents (n=191) indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. Of these the MSP-RETA evaluation was named by six respondents. Similarly, of the 44 respondents naming a publication, four named a publication associated with MSP-RETA. Finally, in response to the question regarding

influential STEM evaluators, evaluators associated MSP-RETA were not mentioned in the 59 responses.

## Cross-Case Analyses

As mentioned earlier, this cross-case analysis is focused on the relationship of project involvement in evaluation efforts and the use of evaluation processes and outcomes by the projects and the fields of STEM evaluation and education. Several themes emerged from analyses across the cases: perceptions of evaluator quality, interface with NSF (requirements, cooperation and outreach), life cycles (program, project and individual levels), project control, community building and networking, and tensions. The themes are examined below by applying the lenses of involvement and use by both the local projects involved in the evaluations and the fields of evaluation and STEM education.

**Perceptions of Evaluation Quality**

Project leaders' perceptions of evaluation quality emerged as a theme linking involvement and use. Projects tended to report more use of the evaluation processes and outcomes when they were involved with what they viewed as high quality evaluation efforts. The four evaluation projects were similar in their capacity to conduct evaluations. All the evaluations were funded on the basis of proposals that went through a review process that favors well-designed implementation plans. They were all funded at appropriate levels to conduct the activities they had planned. All four eventually had money remaining in their budget at the end of the projects to roll over into no cost extensions. All of the project PIs were distinguished, experienced and well-known STEM evaluation specialists with expertise in both STEM and evaluation. Three of the four PIs remained with the project from beginning to end. Only the MSP-RETA experienced a change in leadership as a result of a retirement and a transition of the second in command to the leadership role.

All four of the program evaluation projects used surveys and most used observations and interviews as well. In all but the MSP-RETA case, much of the data collected at the project level could be aggregated to the program level. Even in the MSP case, some information about the projects was aggregated. All four produced high quality evaluation results that were valued by the individual projects at least to some extent. There is some evidence of use of the products of the evaluations by the field such as the DIO model from the MSP-RETA, the methodology brochures developed by ATE, and especially the instruments developed in the LSC and CETP evaluation projects.

This theme suggests that regardless of the amount or type of involvement, project staff tend to use processes and products more if they view the evaluators as well-qualified. Use by the field also seems to be affected by the reputation of the quality of the evaluation. This appears to be related to dissemination in terms of how the evaluation outcomes are 'packaged' and the need

for the evaluation results by the field—how relevant the findings are to important issues in the field.

**Interface with NSF**

How the evaluation projects interacted with NSF affected the relationship between involvement of the projects and the projects' use of the evaluation processes and outcomes. Funding for all four of these program evaluation projects was directly provided by the NSF rather than from the individual projects that were part of the four multi-site evaluation efforts. Likewise, all four program evaluation projects had NSF as the primary audience for their findings and functioned within the context of NSF. One relevant contextual factor is that NSF is unique as a federal agency because it is completely grant oriented with no block grant programs as seen in the Department of Education or line research departments such as NASA or NOAA. In addition, NSF awards its funds annually and is staffed by professionals who choose to work at NSF for 1-3 years and typically return to their previous jobs. Also, as is true for all government agencies, NSF and its programming is subject to changing political forces. Consequently, at different points in time, NSF had different ideas about what involvement might be appropriate by individual projects in multi-site evaluation efforts. How NSF viewed the involvement or cooperation of the individual projects with the evaluation projects varied. In the case of the MSP-RETA, NSF restricted access to the projects. However, NSF was neutral about project involvement for the CETP evaluation. In contrast to both MSP-RETA and CETP, all ATE projects were strongly encouraged to participate, and in the case of LSC, cooperation was required. Thus, the relationship between involvement in and use of evaluation reported generally aligned with the extent of NSF requirements for cooperation.

The interface with NSF also affected the outreach (a type of involvement) by the evaluation efforts to the individual projects and the fields of STEM evaluation and education, which in turn affected awareness and possible use of evaluation processes and outcomes. Because of an emphasis on research, NSF extends the traditional notion that evaluation is knowledge generation to include the idea that receiving and using evaluation results is embedded in dissemination to the field. In other words, dissemination is viewed as a type of involvement. NSF's emphasis on dissemination to the field varied across the four projects and over time. For example, the LSC evaluation project was asked to increase dissemination activity or outreach to the field toward the end of their contract. The ATE evaluation project was encouraged to disseminate to the field soon after the project began. The CETP project wasn't encouraged to disseminate by NSF. The MSP-RETA project was given extra funds to enhance dissemination efforts by conducting national workshops. These differing NSF emphases led to varying interest in dissemination and use of different dissemination techniques by the evaluation projects. Of course, each evaluation project had its own motivations for outreach that were not necessarily related to NSF. Most salient was the desire to publish articles, which is generally encouraged by NSF, but was not specifically encouraged in the case of these evaluation projects.

Despite the NSF notion that dissemination of knowledge results in use, the relationship between dissemination of evaluation outcomes and use is mixed. The ATE program had the most diverse and extensive dissemination strategies, but these efforts did not result in the most extensive use. The LSC had the most evidence of use but had much more traditional and less diverse dissemination strategies. One of the more prominent methods of outreach for the evaluation projects was the dissemination of findings through presentations at PI meetings, which are attended by the individual projects and sometimes projects from other NSF programs. Thus, outreach and dissemination are not the same as more direct involvement in the evaluation processes but do serve as a type of involvement in multi-site settings.

These results show that outreach and dissemination should be considered a form of involvement for both the projects and the field. In addition, projects have other more intensive opportunities for involvement. The relationship with NSF affects both involvement and use. Projects can view the program evaluation project's relationship with NSF as a conduit for them. Projects also are more willing to be involved with and make use of evaluation processes and outcomes when they believe NSF supports the evaluation effort. NSF supports dissemination, which leads to more involvement with the field. This awareness is the first step in using the evaluation outcomes. In addition, if the field believes NSF values the results of the evaluation the field is more likely to use them.

**Life Cycles**

Examination of all of the program evaluation projects showed that the relationships between involvement and use were related to the various life cycles of the projects. These life cycles were experienced at three levels: the program evaluation project, the individual project, and the individuals within the projects. Each program evaluation project had a unique life cycle, which varied according to the timing of the evaluation in relation to the program itself. In addition, the individual projects responded differently to the evaluation depending on where the individual project was in its own life cycle (e.g., final year of funding or a busy time of the year for the project). Lastly, life cycles also existed at the individual level where people at the project level had different experiences and remained or left the projects during the course of the program evaluation.

At the program evaluation project level, there were differences between when the evaluation project began in relation to when the respective program began. For example, the ATE evaluation project began several years after the ATE program had begun and both are still in operation. The CETP evaluation project began after the CETP projects had been in operation for some time and ended after the CETP program was discontinued. The LSC evaluation project began almost simultaneously with the LSC program and continued after the program had been discontinued. The MSP-RETA evaluation project began with the MSP program but was discontinued before the MSP program ended. These different life cycles led to different patterns of use and involvement. In ATE, the projects continue to participate in the annual survey, but were involved only peripherally in the other aspects of the program evaluation

project. Involvement in and use of the ATE evaluation appeared to be somewhat unrelated since all projects essentially participated in the survey and only some report using the evaluation project data. In CETP, the projects were much less concerned about the program evaluation because the actual CETP program had already been discontinued. Involvement in the program evaluation enhanced the provision of data to the program evaluation but seemed to be only slightly related to use of evaluation project data. The relationship between involvement and use appeared to be mediated by the development of a community of CETP project evaluators facilitated by the evaluation project. Participation in the LSC evaluation project was essentially required after the first few years. This enforced involvement appears to be related to more use of the evaluation project outcomes and processes than in the other three evaluation projects. Also, the continuation of the LSC evaluation after the discontinuation of the LSC program allowed for consolidation of the findings and potential implications of the implementation of the program. The LSC evaluation produced many reports and summaries that were convenient for the field to use when applying for new grant projects or in producing their own publications. As discussed above, the MSP-RETA program evaluation effort differed from the other three in that it was an evaluation technical assistance project rather than an actual program evaluation. The life cycle of this project started at the beginning of the MSP program but stopped before the overall program was discontinued. This allowed the MSP-RETA project to provide a forum for discussion among the various MSP projects and to provide information on best practices.

The project level life cycle implications for involvement and use overlapped in various ways with the program evaluation project life cycle. In ATE, all projects have been involved in taking the survey after their first year of funding. This involvement through participation in the survey, however, does not appear to have produced uniform use of ATE evaluation project processes or products. In CETP, the projects that were more mature when the program evaluation began were less receptive to the evaluation because they had their own local evaluations already in place. The newer CETP projects felt that the CETP program evaluation was useful because it had instruments and evaluation plans available that they could use as they were beginning. The involvement in and use of CETP program evaluation processes and products appeared to be mostly related to individual project characteristics. Some reported relationships between involvement and use while others did not. After the first years, all LSC projects were required to participate in the LSC evaluation project. This resulted in almost complete involvement and was related to more use of program evaluation processes and outcomes. In the MSP-RETA, individual involvement was notably varied in terms of the specific help with local evaluations but widespread in terms of participation in the conferences. This appeared to result in patterns that showed people being involved in evaluation activities and using processes and results that they felt were important to their projects.

The individual life cycle implications were, as would be expected, more unique. Each person in all of the individual projects throughout the four evaluation projects was following his or her own path. This resulted in diverse patterns of involvement and use by individuals (versus projects). For example, a few people who were initially involved eventually left their projects

and, therefore, seemingly made less use of the evaluation project processes and results. However, because those individuals left, it is difficult to say how the involvement might affect the individual's long term and personal use of knowledge, skills, and beliefs they may have acquired during their participation in the evaluation projects. One brief example is that of a person involved in one of the evaluation projects who became an NSF program officer. Although, of course, this person had unique personal qualities, it is certainly possible that this person's involvement in one of the evaluation projects had influenced how that person's knowledge, skills and beliefs evolved.

The intersecting life cycles of the program evaluation project, the local projects and the personnel at the local projects affects involvement and use. Likewise, the life cycle of the evaluation project and individuals affect involvement and use by the field. Projects are more easily involved in evaluation efforts at different stages but especially at the beginning when other goals are not in conflict with the goals of the evaluation. This involvement can lead to more use. The field is more likely to be involved with and use evaluation outcomes at the end of a program evaluation project when the data are more complete and dissemination of the lessons learned can be stated more definitively and with more power

**Project Control**

The amount of control a local project had over its involvement in the program evaluation was related to the relationship between involvement in the program evaluation and use of the program evaluation processes and products. In terms of involvement, the MSP-RETA evaluation allowed the individual projects the most control, although this often meant that the projects only minimally interacted with the MSP-RETA. The other cases also varied in the amount of control the projects could exercise over the evaluation, with the ATE case providing the least control, followed by the LSC case and then the CETP case. How the amount of control affected the interaction of involvement and use is not clear. It appears that absolute control, as in the MSP-RETA, does result in use of the advice and information provided. However, the advice most used was that provided directly to the projects by experts. The choice of the projects to attend MSP-RETA meetings was more generalized and appeared to result in generalized increases in knowledge. In the CETP case, the involvement was voluntary, but those who chose to participate seemed to grow in knowledge and skill. This suggests that voluntary involvement capitalizes on individual or project need and results in a 'just in time' learning situation, which can be beneficial. The LSC required involvement so the projects and their evaluators had to complete certain evaluative activities. Although it appears that sometimes this requirement was onerous for local project staff, it did contribute to more use. In the ATE evaluation project, there were different elements that allowed different levels of project control. It appears that although the most involvement was related to the completion of the annual survey, the differing levels of control within different project activities resulted in diverse use of the evaluation project information.

The effect of local project control on involvement in and use of evaluation processes and outcomes is nuanced. Requiring participation, of course, results in involvement but this

enforced involvement doesn't necessarily result in more use of processes or outcomes. Complete local control results in less involvement because projects are too busy to be involved if the choice is left up to them. However, if they choose to be involved, use is likely because it is tailored directly to the needs of the project. The approach that seems to result in the most involvement and use is one that requires participation but also allows the local project to gather and interpret their own data coupled with the opportunity for feedback to the evaluation project.

**Community and Networking**

This theme shares similarities with ideas already raised regarding interface with NSF. However, the focus here is on the evaluation project context rather than the NSF. This theme emerged because it is often cited or inferred that involvement in a community of practice or network creates the sense that people are part of a community and share in the benefit of the group's efforts. In that sense, networking allows the identification and verification of different skills and capacities and the feeling of being part of a team. Team members feel more obligations to work with the team and feel part of what the team produces. This could lead to more use of the processes and products. All but one of the evaluation projects showed clear evidence of the development of community and networking among the individual projects and the exception still had at least some evidence of networking. There were also indications about how networking and community building was developed. One aspect of the development was that the evaluation project PIs truly wanted input from the projects and in order to get the input they needed to interact with the projects. The projects indicated a desire to interact with the evaluation PIs because of the PIs' strong reputations with NSF and their skills in both STEM and evaluation. All of the four evaluation projects had experienced leaders and strong staff. Evidence from this study supported the value of this expertise in building community and networking.

The CETP, LSC, and MSP-RETA evaluation projects all hosted meetings of their own to discuss various aspects of the evaluation project or to provide technical assistance opportunities. In contrast to the NSF PI meetings, where evaluation was only one of the topics, these evaluation projects sponsored meetings that were focused completely on evaluation. These smaller meetings facilitated the development of personal relationships among the projects and between the individual projects and the evaluation project. The LSC project hosted meetings to discuss use of the evaluation instruments and incorporated the comments at these meetings into future training materials. This resulted in higher feelings of involvement and in perceived growth in knowledge and skill and perhaps in other uses as well. The CETP meetings were designed to provide input into evaluation planning and interpretation of data. These did result in feelings of involvement and belief in increased knowledge, but did not appear to be related to instrumental use of evaluation products. The MSP-RETA was particularly singled out as providing increased understanding of evaluation. The PI of the MSP-RETA was very open and supportive and tried to provide the MSPs with the expertise they needed without trying to dominate. The MSP-RETA itself was designed to provide technical assistance so it would be

expected that projects would view it as providing expert advice. In addition to the meetings sponsored by the MSP-RETA, it provided workshops led by national experts for the participants.

The ATE evaluation project was the only project that did not host evaluation-related meetings, and therefore local projects did not associate a sense of community among the local projects with the evaluation, as was the case in the other three projects. However, the ATE evaluation project consistently made both formal and poster presentations at the annual PI meetings. This also involved meeting and talking with individual project members and exchanging contacts for further communication. The ATE evaluation project also made extensive use of the web to present its data to the ATE projects and the fields of STEM evaluation and education. The web site showed significant use and many project personnel talked in depth with the ATE evaluation project staff at the meetings or later online. Consequently, it appears that there may have been some community building, just not as clearly as in the other cases.

In addition to attending and presenting at PI meetings, the culture within the four evaluation projects was to acknowledge the expertise of both the evaluation project personnel and the project personnel. For example, the efforts of the MSP-RETA are highlighted above. The ATE web site allowed individual project PIs to conduct their own analyses and share the results. The LSC used subsets of projects to help make decisions and made changes in training materials based on feedback from participants. The CETP hosted a listserv where the individual projects could post questions and receive help from each other. These cultures of sharing expanded the expertise inherent in the evaluation projects to the entire group of projects; thereby producing a network and a community, rather than a top-down knowledge provision system. The relationship between development of community and use of evaluation products, however, is not direct. Development of community does not necessarily result in use of the evaluation processes and products.

An interesting interaction of involvement in and use of evaluation emerged from the data. The involvement of the individual projects with the evaluation projects influenced NSF. In all four evaluation projects, there was a tendency for the evaluation projects to act as liaisons between NSF and the projects, which helped to develop a sense of community between the projects and the evaluations. The evaluation projects could present consensus opinions from the local projects concerning evaluation to NSF. NSF in turn used the evaluation information to make decisions, which benefitted the projects. This was especially true in the CETP evaluation project where the projects felt they needed the evaluation project to meet NSF needs. The evaluation project could speak with the force of the community rather than the individual projects negotiating with NSF separately. The MSP-RETA used its community of projects to develop an approach to evaluation, the DIO model that met the needs of NSF and allowed the community to have a common language. In the LSC, the community provided definitions of quality in terms of STEM lessons in particular that the projects would then use in their discussions with NSF. The LSC and ATE evaluation projects, as direct information providers to NSF, helped to move the LSC and ATE programs forward in the directions represented by the individual projects.

Development of a community appears to have fostered both involvement in the evaluation project and use of evaluation process and outcomes. The more community building efforts were supported, the more involved the projects felt. At least some of the community building efforts need to be face-to-face and focused on the evaluation for community building to occur. There also needs to be clear evidence of mutual respect. Development of community may be peripherally related to involvement and use by the field as projects that are part of the community bring in other members from the field.

**Tensions**

There was evidence in all four projects of tension between different alternatives; a sort of double edged sword effect related to resources. These tensions affected involvement, and possibly use, negatively. In general, although individual projects valued the development of community and networking offered by the evaluations, projects resisted participating because of the constraint it placed on resources like time and money. Several ATE projects mentioned that the survey took too much time, yet they knew it was important to complete it. In CETP, participation by the individual projects in the evaluation efforts was inconsistent because of the amount of time involved in collecting the core evaluation data. In LSC everyone participated, but people felt that the time might have been better spent doing other things. In MSP-RETA the evaluation consultants were not used as much as had been allocated, even though the projects had expressed a need for such services. In the LSC all projects, big or small, had the same core evaluation costs. This was a significant burden for the smaller projects. The tension focused on where the smaller projects could get the money to support activities in light of the high evaluation costs. In the ATE and CETP evaluations there were external problems with funding that resulted in delayed or changed activities and created tension about how to decide what to do when funding was not available.

Another critical tension was the balance between understanding the need for the national evaluation projects, while wanting to prioritize the local individual project evaluations because it had a more direct effect on the project. The lines between these two endeavors were blurred for many respondents. Additionally, local evaluation efforts were designed to highlight the unique character of the individual project; whereas the national evaluation projects were designed to highlight the character of the program overall. Many individual projects across the four cases expressed frustration that the national evaluation examined the lowest common denominator and glossed over the uniqueness of their project. The tension was increased because the projects were not the directly intended recipients of the evaluation projects' efforts, so the evaluations were not necessarily designed to meet their needs. This brought up questions of why they were involved and what they were supposed to do with the results.

Tensions between the project goals and the evaluation goals affected involvement and perhaps affected use as well. From the projects' perspectives, the evaluation might not be as well aligned with their goals, therefore they were reluctant to participate and were skeptical about the validity of the results. There was little evidence that these tensions affected involvement

and use by the field although low project opinion may have discredited evaluation outcomes for those connected to the projects in some way.

**Unique Aspects**

Although similarities across the cases are discussed above, each case was also unique in its relationships between involvement and use. In CETP involvement produced positive feelings and a community of evaluators. This involvement resulted in conceptual use (in terms of knowledge, beliefs and skills), but it did not appear to be related to instrumental or broad field use of the evaluation products. In LSC, the extensive involvement of the projects, the required discussions between the local project evaluators and PIs, and the monitoring of the evaluation data provided by the projects to the core evaluation team appeared to increase both the conceptual and the instrumental use of the evaluation results by the projects and the field. It also appears that there was more use of the evaluation publications. Thus, this arguably mandatory use was associated with greater project-level engagement with data collection instruments, protocols and data, and greater interest in the overall results using theirs and others' data. In ATE, despite extensive dissemination efforts on the part of the evaluation project, local project staff did not report using or being influenced very much by the overall program data. Instead, the projects reported feeling involved often because of filling out the survey. In the MSP, the relationship was much more complex because of the different possible levels of participation; actually having an evaluation consultant work with the project versus attending a workshop. As would be expected, there appeared to be more instrumental and conceptual use when there was individual evaluator work although the workshops appeared to result in conceptual use mostly influencing dispositions toward evaluation.

# Conclusions

The relationship between involvement in an evaluation project and use of that project's processes and outcomes is complex. This complexity is exacerbated by the fact that the involvement and use being examined is by persons who are not the directly intended users of the evaluation. Furthermore, there were two types of unintended users included, the projects who participated in the evaluation project and the field of STEM education and evaluation.

Involvement by the local projects in the evaluation project appears to be only one aspect promoting use of evaluation project processes and outcomes. It appears to be an enabler in some instances but not necessarily a strong or consistent one. Naturally, project personnel would have to know something about an evaluation to make use of it, but the pathway between knowing of it and using it appears to be circuitous. Face-to-face presentations and other forms of dissemination and outreach are necessary precursors to involvement, and involvement would provide information about the evaluation processes and products.

Development of community appears to be a stronger force than simple involvement, although involvement could lead to development of community.

Involvement by the field is perforce less intensive. In fact, most involvement by the field tends to be embedded in the evaluation project's outreach and dissemination efforts, although some involvement is through the projects. Use by the field appears to be promoted by close relationships of the evaluation to NSF, by the evaluation project providing instruments that could be used in other settings and by the evaluation project providing easily used or referenced summary materials. Overall, it appears that these four evaluation projects may have had little impact on the fields of STEM education and evaluation. This may be due to limitations in our ability to measure the effects. It may also be because in both STEM education and evaluation, information from one study is not tightly built upon in future studies such as is the case in the bench sciences. Perhaps it is because there are too many variables to consider simultaneously and there are individual motivational and other interaction effects. Also, evaluation results are more often provided to and used by the directly intended user (NSF in this case) rather than the unintended user (the field in this case). More synthesis of findings from diverse studies and targeted use studies would help. If NSF wants to encourage this, the agency could conduct more syntheses of evaluation and research findings and incorporate this information into requests for future proposals.

The results of this cross case study suggest several ideas for affecting involvement and subsequent use of evaluation processes and outcomes by unintended audiences. The most important aspect of facilitating involvement and use appears to be careful and continuous communication by the evaluation to the participating projects as well as the field through a variety of mediums. It is important to have a clear interface with the direct user of the evaluations because this relationship affects how the individual projects and the field interact with the evaluation. Evaluators need to be cognizant of the life cycles of the individuals and the projects involved in the evaluation and plan for their different types of involvement and use. Tensions between the overall evaluation and the local projects and their evaluation efforts should be expected. A seemingly fruitful way to use evaluation funds is to develop a sense of community among the local projects through face-to-face meetings as well as other types of contact. Developing a community of practice with mutual respect and true interest in each other's ideas as well as reasonable expectations for participation are important.

Across all of the cases, the perceptions of involvement by the individuals in the projects were not consistent. Perceptions of degree of involvement varied by type of activity engaged in, and by the demand of the activity. Additionally, individuals had their own idiosyncratic, and not necessarily consistent, standards for what it meant to be involved a lot or a little. Various involvement schemas have been proposed in the evaluation literature. The four cases confirm some parts of those schemas and also provide additional aspects (Toal, et al, 2009). In general, the literature considers involvement at different points in the evolution of an evaluation and considers involvement as including aspects of control over the evaluation design, implementation or interpretation/communication of results, e.g., helping to design evaluation

questions, helping to interpret data. (Cousins, 2003) Our data show, however, that at least two other activities—providing data and attending meetings—were also viewed as involvement.

This cross-case analysis has several limitations. It represents only four instances of large, multi-site NSF evaluations and therefore generalizations to other settings are not possible, although potentialities can be suggested. The case studies themselves are based on self report data along with some archival records. The numbers of people surveyed and interviewed are small but appear to be at least representative of the groups included. The instruments used for data gathering were developed as part of the project and therefore might not be as valid measures of involvement and use in other contexts.

This research project is unique in that it essentially begins the study of evaluation use and involvement by unintended users. As described, the four evaluation projects were funded by NSF and NSF was the direct user in all cases. Despite this, individual projects and the fields of STEM education and evaluation were also important. Their use of the evaluation processes and outcomes could be considered a bonus and a glimpse down the path to the broader types of societal use suggested by Mark and Henry (2003). This research study had to develop new methods to ascertain the effect of involvement in and use of evaluation on unintended users such as projects (e.g., surveys of involvement and use) and fields (network analysis of citations and surveys of PIs and editors) as opposed to effects on individuals.

Future work should include more research on the causal nature of involvement with evaluation use. The themes presented here provide fruitful areas for more investigation. Mark and Henry (2003) called for more theoretical and aligned work in evaluation so that studies can build more effectively on each other. This cross-case analysis provides a strong baseline for more positivistic research. It might be possible to examine the issues raised here through more quantitative path analytic procedures. It is also important to develop strong theories about the relationship between involvement and use that could form the basis for hypothesis formulation and subsequent research.

In summary, it appears that perceptions of involvement in these four multi-site evaluation projects are different from the perceptions considered in traditional participatory evaluations. This is partially true because of the different levels of participants in a multi-site evaluation but also because of the different nature of the type of evaluation being conducted. In a multi-site evaluation the participant is really a site or project rather than an individual, but of course it is individuals at the project level that react (Lawrenz and Huffman, 2003). These project-level participants are generally members of the project leadership as opposed to participants in the project, which would be the case in more traditional participatory evaluations. Additionally, the projects and the field were not the intended users of the evaluations; NSF was. It is also clear that individuals themselves have quite diverse opinions of what constitutes involvement. They perceive involvement from a different perspective than evaluation theorists such as Burke (1998). The participants studied here were differentially affected by the depth and breadth of involvement in evaluation activities. Neither breadth nor depth was consistently predictive of

perceived level of involvement. This lack of consistency in perceived involvement makes measuring involvement challenging. A similar case could be made for perceived use. This suggests that any investigation of the relationship between involvement and use will require in-depth measurement work and is likely to be substantially affected by the nature of the evaluation and the characteristics of the individual.

| THEME | ATE | CETP | LSC | MSP-RETA |
|---|---|---|---|---|
| **Lifecycles: Program-evaluation level** | • The evaluation team invited input from different groups at different times. | • The CETP Core Evaluation was implemented after many projects had already been funded and had their own project-level evaluations in place. | • The Program evaluation was new, so some projects were reluctant to participate due to unfamiliarity.<br>• Timing of funding affects use. | • There were differences in timing—the RETA was funded initially with a few MSPs but then other MSPs were added. |
| **Lifecycles: Project-evaluation level** | • Project perceptions of the annual survey changed during the project's duration. | • "Mature" CETPs and "developing" CETPs had different experiences. | • Increased conflict between the projects and the Core when the project evaluations pre-date core evaluation.<br>• More impact on field after program done. | • Both the MSPs and the RETA went through significant evolution.<br>• Need for working from the start to the end of MSPs' evaluation projects. |
| **Lifecycles: Individual level** | | • Previous evaluation experience.<br>• Other individual factors (retirements, moves, etc.). | • Experience level of local evaluations affects attitude toward Core.<br>• Taking the long view on give-and-take affects attitude toward involvement. | • Previous evaluation experience. |
| **Project control** | • Project-Level Control: The evaluation team controlled the evaluation.<br>• Program-Level Control: NSF offered input, not instruction. | • Participation in all CETP core evaluation activities was strictly voluntary. | • Funding mechanism prescribed level of control.<br>• Involvement fostered ownership. | • No intended control/requirements to participate.<br>• Emphasis on rigorous evaluation<br>• Some projects felt that they were required to participate. |

| THEME | ATE | CETP | LSC | MSP |
|---|---|---|---|---|
| **Perceptions of Evaluation Quality** | • Experienced evaluator | • Experienced STEM evaluator | • Experienced STEM evaluator | • Experienced evaluator |
| **Interface with NSF** | • All projects strongly encouraged to participate in evaluation project<br>• NSF encouraged evaluation project to begin dissemination soon after evaluation began | • NSF maintained a neutral position with regards to local project involvement<br>• NSF did not encourage dissemination as part of the evaluation project efforts | • Cooperation with program evaluation project was required<br>• NSF asked evaluation project to increase dissemination efforts | • NSF restricted access of evaluation project to local projects<br>• NSF provided evaluation project with additional funds to enhance dissemination efforts |
| **Community &Networking** | • Community building was not an explicit desired outcome of this evaluation project<br>• Attended PI Meetings<br>• Hosted a website with local project data and resources | • Attended NSF PI Meetings<br>• Hosted meetings to solicit input on evaluation planning and interpretation of data from local projects<br>• Used a listserv for local projects | • Attended NSF PI Meetings<br>• Hosted meetings with local projects to discuss use of evaluation instruments<br>• Used subsets of projects to make decisions and changes in training materials | • Attended NSF PI Meetings<br>• Hosted meetings to build evaluation capacity of local projects<br>• Provided technical assistance through a network of experts |

| THEME | ATE | CETP | LSC | MSP |
|---|---|---|---|---|
| **Tensions** | • Too much time required.<br>• Boundaries between national and local project evaluations.<br>• Local highlight unique features vs. national focus on common features. | • Too much time required.<br>• Boundaries between national and local project evaluations.<br>• Local highlight unique features vs. national focus on common features. | • Too much time required.<br>• Cost significant burden, especially for smaller projects.<br>• Boundaries between national and local project evaluations.<br>• Local highlight unique features vs. national focus on common features. | • Too much time required.<br>• Boundaries between national and local project evaluations.<br>• Local highlight unique features vs. national focus on common features. |
| **Unique Aspects** | • Local project staff did not report using or being influenced by overall program data | • Involvement produced positive feelings and a community of evaluators resulting in conceptual use | • The extensive involvement appeared to increase both conceptual and instrumental use | • When the expert consultation was used, there was increased conceptual and instrumental use<br>• Those who attended the workshop reported greater conceptual use |

# References


Burke, B. (1998). Evaluating for a change: Reflections on participatory methodology. *New Directions for Evaluation,* (80), 43-56.

Cousins, J. B. (2003). Utilization effects of participatory evaluation. In T. Keleghan, & D. L. Stufflebeam (Eds.), *International handbook of educational evaluation* (pp. 245-268). Boston: Kluwer Academic Publishers.

Katzenmeyer, C. & Lawrenz, F. (2006). National Science Foundation Perspectives On The Nature Of STEM Program Evaluation. *New Directions For Evaluation,* (109), 7-18.

Lawrenz, F., & Huffman, D. (2003). How can multi-site evaluation be participatory? *American Journal of Evaluation, 24*(4), 331-338.

Mark, M. M., & Henry, G. T. (2004). The mechanisms and outcomes of evaluation influence. *Evaluation, 10*(1), 35-57.

Toal, S. A., King, J.A., Johnson, K., & Lawrenz, F. (2009). The unique character of involvement in multi-site evaluation settings. *Evaluation and Program Planning* (32)*, 91-98.*

Toal, S. A. (In press). The validation of the evaluation involvement scale for use in multisite settings. *American Journal of Evaluation.*